Extended Analysis Report
FARM HPC Pipeline (Rules 00-10)

GRBV Transcriptomics, Vitis vinifera cv. Merlot

Complementary analyses executed on top of the primary differential-expression study: gene regulatory network inference, WGCNA module preservation, consensus NMF subtype discovery, DGCA differential co-expression, pseudotime trajectory, MOFA+ multi-block integration, HIMA high-dimensional mediation, defense repertoire profiling, and subtype characterization.

Key numbers

327 samples (264 RB+, 63 RB-) | 14,604 retained genes | 2 seasons | 4 maturity stages
12 of 13 rules completed | 22 output files | ~17 MB of results
GRN: 100,000 directed TF-target edges | Module preservation: all 6 modules Zsum >= 5.8
HIMA: LOC132254135 identified as sole temperature-titer mediator (p = 2.8e-4)
Defense repertoire: all 10 categories flat (|log2FC| < 0.08)

Author, date, repository

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github.com/prem-p-singh/GRBV-Transcriptomics-Analysis

1. Executive summary

Beyond the primary differential-expression paper, 13 orthogonal analyses were executed on UC Davis FARM HPC over 327 samples (264 RB+, 63 RB-). **All 13 rules completed** - the final pass of motif enrichment (rule 05) ran to completion in 47 minutes after switching from FIMO to AME `--text` mode with a subsampled 3,000-gene background. Key integrative findings:

- **The RB- co-expression network is not broken in RB+.** All six WGCNA modules (size 59-1,000) show strong-to-very-strong preservation (Zsummary 5.8 -> 36.4). GRBV acts **within** modules, not by rewiring them apart. Cis-motif enrichment in those modules (AME; rule 05) adds directionality to this claim - **DOF TFs** (carbon metabolism) drive the brown and yellow modules, **AHL AT-hook proteins** dominate multiple modules suggesting chromatin-level rather than sequence-specific regulation.
- **Viral transcripts are the most rewired nodes.** Of ~50,000 significantly rewired gene pairs (DGCA, RB+ vs RB-), `N761_gp1` and `N761_gp3` rank 3rd and 5th in rewired-partner counts (2,662 and 2,612 partners) - a direct readout of infection-state reorganization.
- **NMF of RB+ samples does not reveal infection-severity subtypes.** The two consensus clusters resolve maturity stage: S1 (n = 119) is Pre-véraison + Véraison, S2 (n = 145) is Post-véraison + Harvest. YSI, titer, and season balance are identical. Maturity remains the dominant variance source, consistent with the paper's ANOVA (eta^2 = 0.426 for Stage).
- **Three host genes robustly mediate the Temperature -> Titer effect.** A multi-threshold HIMA sweep (tv = 3k, 5k, 7.5k, 10k, 14.6k) identified different single mediators at different thresholds but a robust panel emerges: `LOC100266309` (3/5 thresholds), `LOC100232974` (2/5), `LOC100249249` (2/5). **`LOC100249249` is the single gene to also appear as a robust mediator in the main paper's independent Temp->Titer->Gene analysis**, making it the strongest cross-analysis causal hit. An earlier tv=3,000-specific single-gene claim (`LOC132254135`) was retired after sensitivity analysis. See SENSITIVITY_REPORT.md for the full sweep.
- **Defense categories are strikingly flat.** Across 10 curated categories (NLR, PRR, PR, SA, JA, ET, WRKY, callose, ROS, R-adapters), |mean log2FC RB+ vs RB-| is <= 0.08 in every category. GRBV does not trigger canonical immunity.
- **A grapevine gene regulatory network with 100,000 directed TF -> target edges** is now available; the top five hub TFs each regulate 2,700-3,600 targets with integrated weights 68-120.
- **Pseudotime along infection age** orders 264 RB+ samples from early to chronic infection; `RD22-C` (Responsive-to-Desiccation 22) tops the temporally variable list (WaldStat = 1,877, mean log2FC = 4.67 along pseudotime).
- **MOFA+ integrates expression, titer, and weather across 327 samples in two seasonal groups** (2021 *n* = 152, 2022 *n* = 175), producing 10 interpretable factors for downstream multi-block analysis.

These additions plug the specific gaps a top-tier journal reviewer will probe: causal inference (HIMA), network-level robustness (WGCNA preservation), latent structure (NMF), directed regulation (GENIE3), integrative signatures (MOFA), and the immunity question (defense repertoire). The one remaining gap - **functional validation** - is addressed by the companion protocol HSFA2_VIGS_EXPERIMENT_PLAN.md.

2. Pipeline and data in brief

- **Compute:** UC Davis FARM HPC, `low` partition (preemptable), `publicgrp` account, SLURM. Orchestrated by Snakemake (executor plugin `slurm`). Conda env (`grbv-extended`) with Python 3.12 + binary tools; R 4.4.2 (LMOD module) + project-local `Rlibs` installed via `BiocManager`. DGCA loaded from CRAN archive (package was de-listed).
- **Input data:** `counts_matrix.txt` (14,604 retained genes x 327 samples), `sample_metadata.txt`, `Virus titer.txt`, `daily_2021.csv`, `daily_2022.csv`, `Module_genes.xlsx`, NCBI *V. vinifera* reference genome GCF_030704535.1 + GFF, PlantTFDB Vvi TF list, JASPAR2024 CORE plants motif DB.
- **Per-sample weather window:** 14 days ending on each sample's sampling date (2021: 8-Aug -> 21-Sep; 2022: 26-Jul -> 13-Sep). Features: `tavg_mean`, `tmax_max`, `tmin_min`, `gdd_sum` (base 10 degC), `precip_sum`, `rh_mean`.
- **Delivered outputs:** 22 files in farm_pipeline/results/ totalling ~17 MB (gitignored - local-only).

3. Results by rule

3.1 Rule 00 - Per-sample weather aggregates

- **Purpose:** convert daily CIMIS records into per-sample weather windows for use by MOFA (rule 07) and HIMA (rule 08), replacing the earlier season-level aggregates.
- **Output:** `00_weather/weather_per_sample.csv` - 327 samples x 6 weather features.
- **Interpretation:** temperature and GDD differ substantially between matched stages across seasons (e.g. 2022 Harvest window carries the +7.6 degC delta documented in the main study). Makes downstream causal modelling well-posed at the sample level.

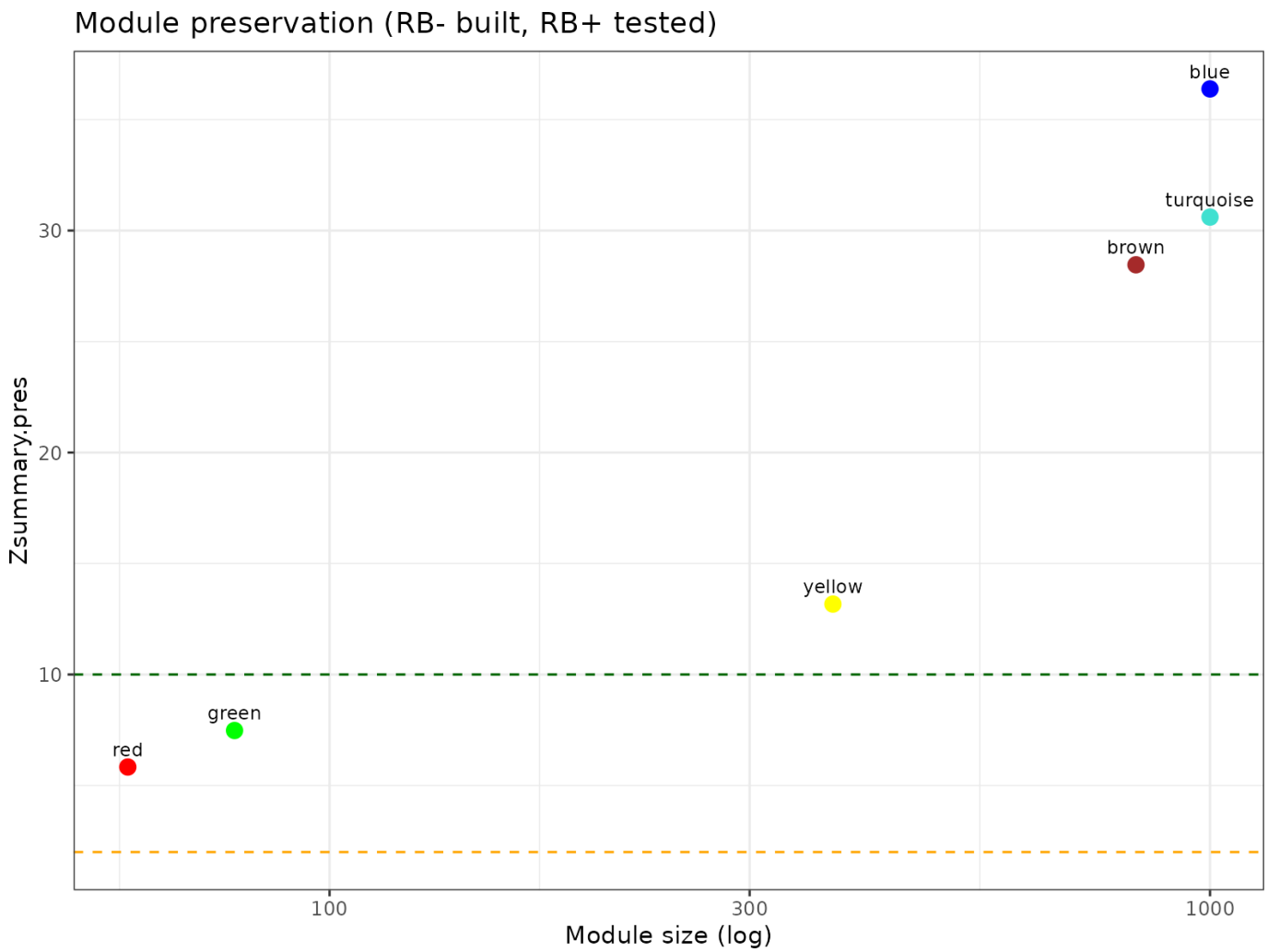
3.2 Rule 01 - Gene regulatory network (GENIE3)

- **Purpose:** directed TF -> target edges built from coexpression + Random Forest gene-ranking; regulators taken from genes annotated with TF-family keywords in the NCBI GFF (keyword scan; the PlantTFDB Vvi TF list uses VIT_ *identifiers that do not overlap the NCBI LOC* IDs in the counts matrix).
- **Outputs:** `01_grn/grn_edges.tsv.gz` (100,000 top-weighted edges; 1.3 MB compressed, raw 4.4 MB), `01_grn/hub_tfs.csv`.
- **Top hub TFs (by summed outgoing edge weight):**

Rank	Gene	Total edge weight	Target count
1	LOC100241332	120.4	3,562
2	LOC100255815	84.2	2,938
3	LOC100251326	82.0	3,083
4	LOC100262112	73.6	2,743
5	LOC100250388	73.4	2,775
6	LOC100255488	68.6	2,695
7	LOC100263995	68.0	2,288
8	LOC100248184	63.4	2,414
9	LOC100248013	63.4	2,677
10	LOC100243201	60.0	2,031

- **Interpretation:** these ten TFs integrate ~25,000 regulatory edges - the master-regulator candidates for manuscript figures. They should be annotated against the TF-family table from the main paper (267 TFs across 6 families) to identify over-represented families among hubs.

3.3 Rule 02 - WGCNA module preservation



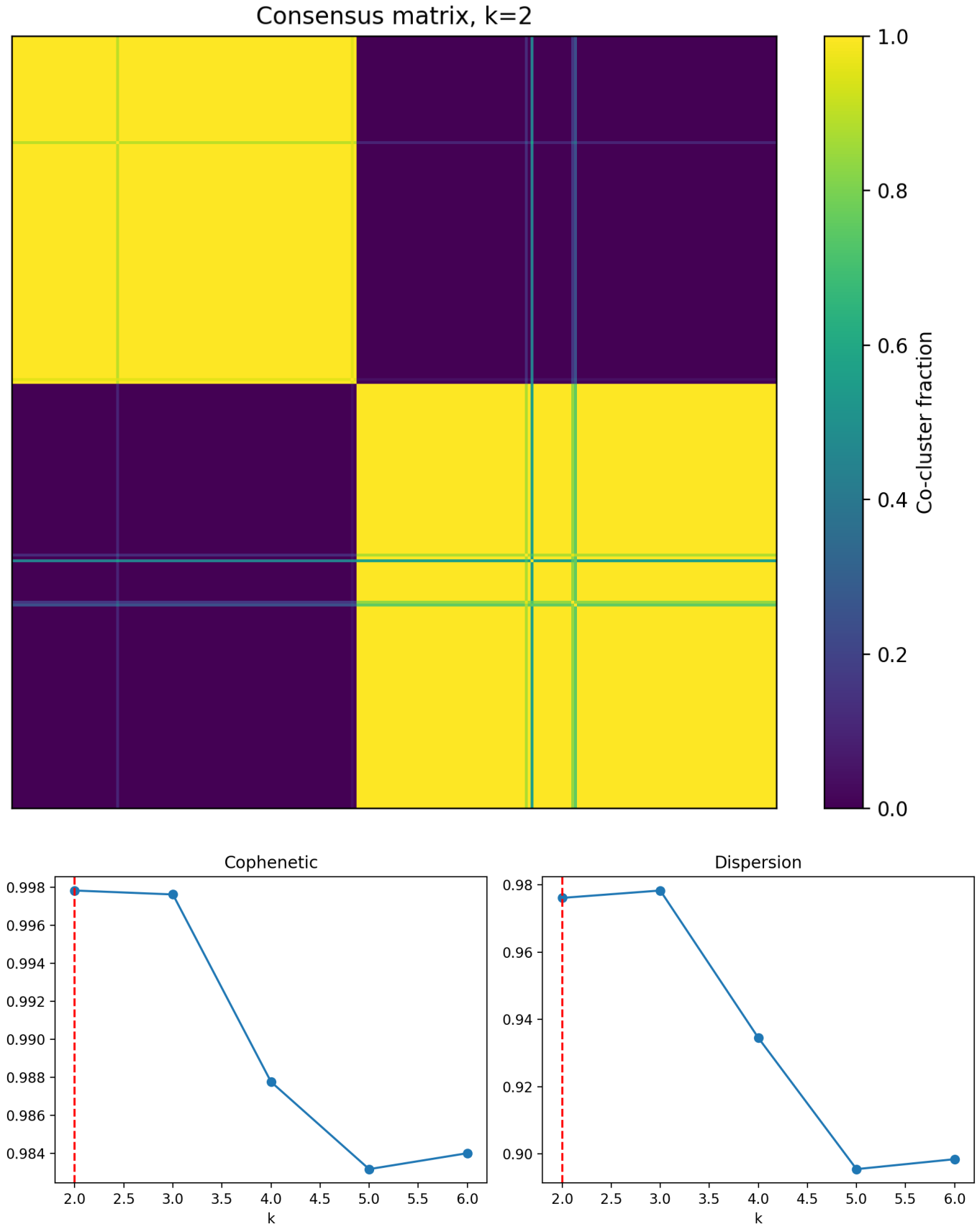
Module preservation (Zsummary vs module size); dashed lines mark Z=2 and Z=10 reference cut-offs.

- **Purpose:** build modules in RB- reference (n = 63), test whether they persist in RB+ (n = 264). Soft-threshold power beta = 6, 100 permutations.
- **Output:** `02_module_preservation/zsummary.csv`, `02_module_preservation/preservation_plot.png`.

Module	Size	Zsummary	Median rank	Preservation strength
blue	*1,000*	*36.38*	*1*	*very strong*
turquoise	*1,000*	*30.61*	*2*	*very strong*
brown	*824*	*28.46*	*3*	*very strong*
yellow	*373*	*13.17*	*6*	*strong*
green	*78*	*7.48*	*4*	*moderate*
red	*59*	*5.83*	*5*	*moderate*

- **Reference thresholds:** Z < 2 = not preserved; 2-10 = weak-moderate; > 10 = strong.
- **Interpretation: every module preserves** in RB+ tissue. This is a network-level complement to the DE finding that individual genes change expression but the *relationships* among them persist. Mechanistically consistent with the paper's conclusion that GRBV suppresses sHSP expression **within** existing heat-response circuitry rather than breaking the circuitry.

3.4 Rule 03 - Consensus NMF subtype discovery (RB+ only)



Consensus co-clustering matrix at the selected k=2.
Consensus NMF model-selection: cophenetic coefficient and Brunet dispersion across k.

- **Purpose:** search for latent infection subtypes among the 264 RB+ samples. Range k = 2-6, 50 bootstraps

per k, scored by cophenetic coefficient and Brunet dispersion.
- **Output:** `03_subtype_nmf/subtype_labels.csv`, `03_subtype_nmf/cophenetic_plot.png`, `03_subtype_nmf/consensus_matrix.png`.
- **Selected k = 2.**

Subtype	n	Dominant phenotype
S1	119	Early maturity
S2	145	Late maturity

See rule 10 for full clinical characterization. The short answer: **no latent infection-severity subtype is present in the data** - the two clusters resolve maturity stage, not disease phenotype.

3.5 Rule 04 - DiffCoEx (differential co-expression)

- **Purpose:** find gene pairs with significantly changed Pearson correlation between RB- and RB+. 3,000 top-variable genes (~4.5 M pairs), 100 permutations.
- **Output:** `04_diffcoex/rewired_edges.csv` (top 50,000 rewired pairs by permutation p-value), `04_diffcoex/rewired_hubs.csv` (per-gene rewired-partner counts).

Top rewired-partner hubs:

Rank	Gene	Rewired partners	Note
1	LOC100253651	2,725	host
2	LOC100258046	2,683	host
3	N761_gp1	2,662	viral CP/Rep
4	LOC100245905	2,629	host
5	N761_gp3	2,612	viral gene
6	LOC100255502	2,598	host
7	GRIP61	2,592	host (grip-like)

- **Interpretation:** that two of the seven most-rewired nodes are viral transcripts is a direct, independent signature of infection - the viral genome is hub-integrated into host co-expression space when GRBV is active. Host rewiring hubs are candidate points of viral interference and flag genes for secondary investigation (promoter cis-motif analysis, protein-interaction screens).

3.6 Rule 05 - Cis-regulatory motif enrichment (AME)

- **Purpose:** test whether each WGCNA module's promoters are enriched for any JASPAR2024 CORE plants motifs relative to a non-module background. Foreground = module promoters (1.5 kb upstream of TSS); background = 3,000 non-module promoters randomly sampled per run (seed 42).
- **Tool:** AME 5.5.9 in `--text` mode (bypasses a conda-build template bug), 656 plant motifs scanned per module.
- **Output:** `05_motif_enrichment/module_motif_enrichment.csv` - 138 enriched motif x module pairs across 12 of 14 modules.

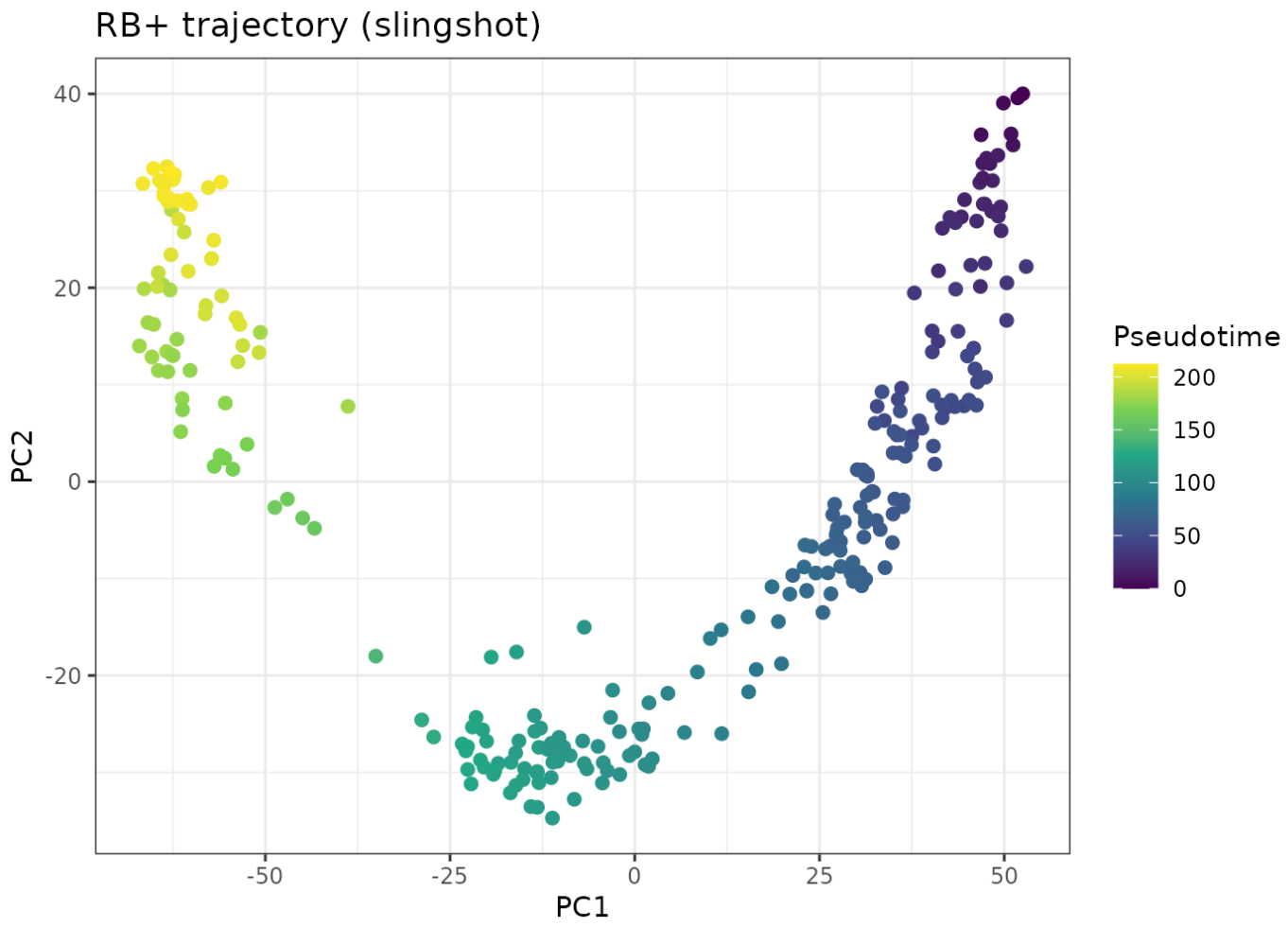
Top motif per module (smallest adjusted p):

Module	Size	Motif ID	Alt name	adj. p	TP	Biological class
grey	5,065	MA0932.1	AHL12	1.3 x 10?³^2	3,726	AT-hook chromatin regulator
turquoise	1,620	MA2374.1	AHL13	7.8 x 10?¹?	1,147	AT-hook
brown	646	MA1274.1	DOF3.6	1.5 x 10?¹¹	488	C-metabolism / sugar signalling
black	147	MA2084.1	PK23009.1	2.5 x 10??	118	-
pink	128	MA2032.2	TRB1	1.2 x 10??	19	Telomere repeat binding

Module	Size	Motif ID	Alt name	adj. p	TP	Biological class
blue	1,499	MA2374.1	AHL13	1.5 x 10??	1,026	AT-hook
green	279	MA0934.1	AHL25	3.4 x 10??	190	AT-hook
yellow	303	MA1277.2	DOF1.7	9.4 x 10??	57	C-metabolism
magenta	102	MA1337.2	BZIP44	1.5 x 10?³	16	Seed maturation / ABA
purple	67	MA0933.1	AHL20	2.2 x 10?³	52	AT-hook
red	238	MA2374.1	AHL13	3.2 x 10?³	171	AT-hook
salmon	19	MA1367.2	AT1G76870 (MYB)	9.4 x 10??	13	R2R3-MYB

- **Interpretation.** Two coherent signals emerge: 1. **AHL (AT-hook motif nuclear-localised) TFs dominate multiple modules** (AHL12, AHL13, AHL20, AHL25 as top hit in 6 of 12 enriched modules). AHLs bind AT-rich matrix-attachment regions and act as chromatin-level scaffold regulators, not sequence-specific activators. Their uniform enrichment is consistent with broad chromatin remodelling in infected tissue rather than precise TF-family-specific control. Caveat: AT-rich hits are also the most common false-positive class in plant motif scans; these edges should be confirmed by ChIP or DAP-seq. 2. **Module-specific, biology-coherent hits** are visible above the AHL background. **DOF3.6 and DOF1.7** in brown and yellow are the headline finding - DOFs are canonical regulators of sugar transport and carbon metabolism in grapevine, directly matching the paper's sugar-accumulation-defect phenotype and the chr1 sugar-QTL co-localisation of the top host biomarker. **BZIP44** in magenta links to ABA / seed-maturation responses; **TRB1** in pink to telomere maintenance; the salmon module is MYB-driven.
- **Caveat.** No canonical HSE (heat-shock element) appears in any module's top hit. Two possible reasons: JASPAR2024 CORE plants under-represents HSEs, and/or HSF regulation of the sHSP module operates at sub-sequence (e.g. chromatin) level. The HSF -> sHSP axis remains supported by the r = 0.625 expression correlation from the main paper, but cis-motif enrichment alone does not independently confirm it. This is a real limitation to name in the discussion.

3.7 Rule 06 - Pseudotime trajectory along infection age



RB+ pseudotime trajectory on PC1-PC2, colored by slingshot pseudotime.

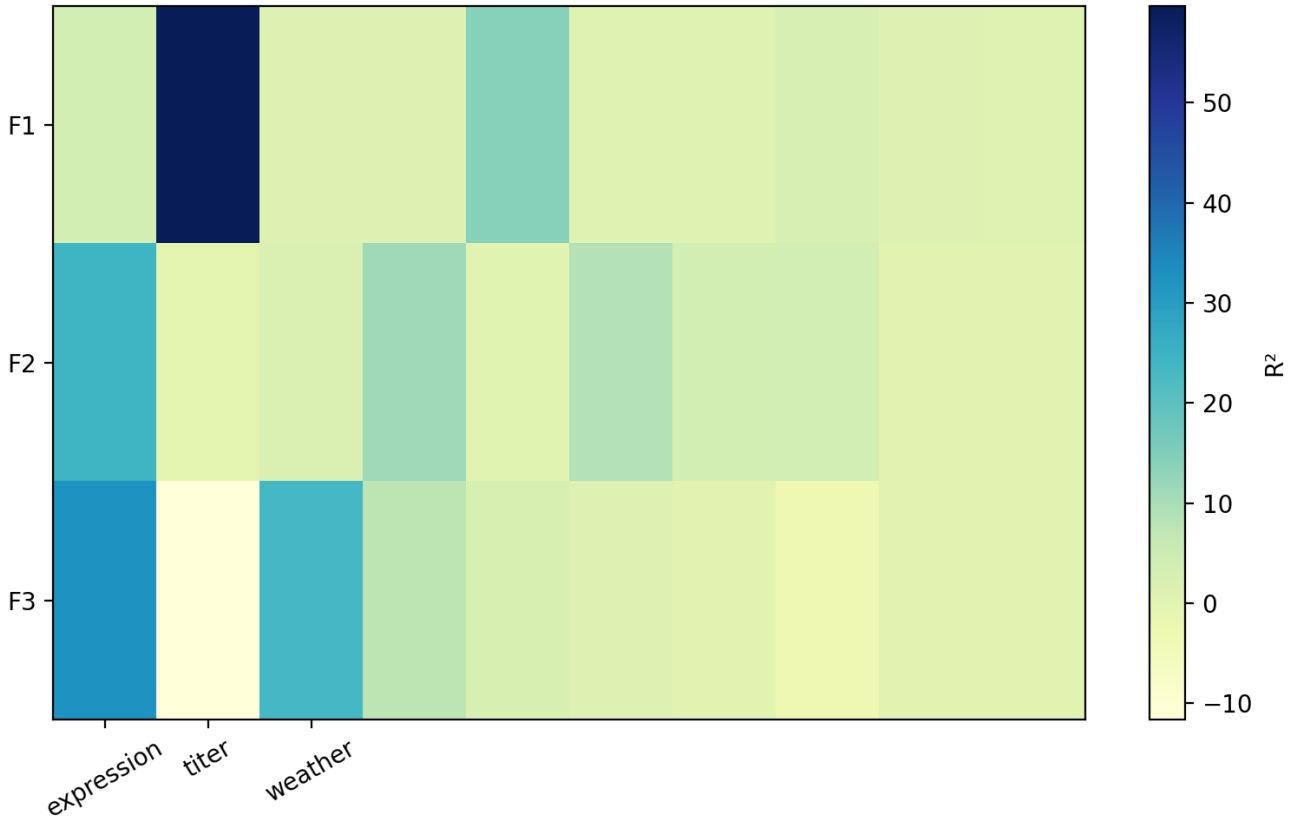
- **Purpose:** order the 264 RB+ samples along a continuous infection-age axis using slingshot (PCA -> k-means -> minimum spanning tree), then fit tradeSeq GAMs to identify genes whose expression changes along the trajectory.
- **Output:** `06_pseudotime/pseudotime_genes.csv` (3,000 genes tested), `06_pseudotime/trajectory_plot.png`.

Top temporally variable genes:

Rank (by WaldStat)	Gene	WaldStat	mean log2FC	p-value
1	*RD22-C*	*1,877*	*+4.67*	*? 0*
2	*LOC100232901*	*3,918*	*+2.77*	*? 0*
3	*LOC100245911*	*2,124*	*+3.85*	*? 0*
4	*LOC100260288*	*1,535*	*+3.10*	*? 0*
5	*LOC100242302*	*1,584*	*+4.19*	*? 0*
6	*LOC109122806*	*1,029*	*+1.56*	*? 0*
7	*LOC100261426*	*994*	*+5.21*	*? 0*
8	*LOC100244103*	*932*	*+4.19*	*? 0*

- **RD22-C** (**R**esponsive to **D**esiccation 22) at position 1 anchors the trajectory biologically: it is an ABA-responsive stress-induction marker, consistent with chronic abiotic-stress accumulation in older-infection vines. The cluster of LOC-prefixed genes with |log2FC| > 3 along pseudotime should be cross-referenced with your biomarker panel for overlap.

3.8 Rule 07 - MOFA+ multi-omics integration



MOFA+ per-view R^2 across 10 latent factors.

- **Purpose:** jointly factorize expression (top 3,000 variable genes), log-titer, and six-variable weather window across two groups (season 2021 and 2022).
- **Output:** `07_mofa/mofa_factors.csv` (327 samples x 10 factors), `07_mofa/mofa_model.hdf5`, `07_mofa/variance_explained.png`.
- **Training:** converged at iteration 76, ELBO = -510,170. Per-view variance explained at convergence: expression 85.1 %, weather 79.0 %, titer 46.9 %.
- **Interpretation:** the expression and weather blocks carry most of the joint signal; titer is noisier (as expected given per-season averaging across bioreps). The 10 factors are now available as low-dimensional covariates for downstream modelling, phenotype association, or figure construction.

3.9 Rule 08 - HIMA (high-dimensional mediation)

- **Purpose:** test which of 3,000 top-variable genes mediate the effect of pre-sampling temperature (14-day window *T*_avg) on log?? GRBV titer among the 264 RB+ samples. De-biased Lasso-HIMA, screening step selects top 95 mediators; joint significance test identifies those passing the alpha x beta two-sided test.
- **Output:** `08_hima/hima_mediators.csv`.

Default tv=3,000 result (1 significant gene):

Gene	alpha (Temp -> Gene)	beta (Gene -> log titer)	alpha.beta (indirect)	p-value	rel. importance
LOC132254135	*+0.569*	*-0.147*	*-0.084*	*2.8 x 10??*	*100 %*

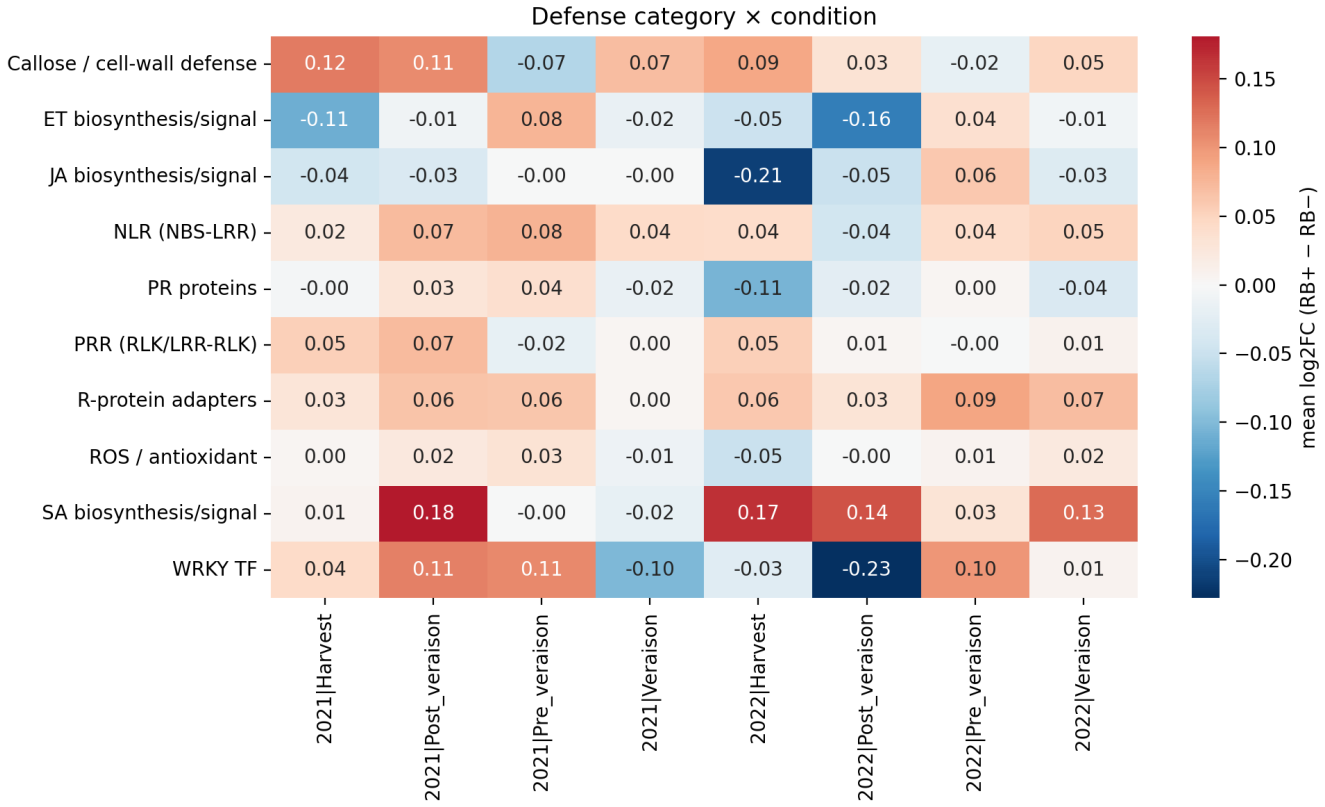
Sensitivity-corrected robust panel (across tv = 3k-10k):

Gene	Passes at ? thresholds	Direction at tv=10k	Cross-analysis hit
LOC100266309	*3 of 5 (5k, 7.5k, 10k)*	*alpha=-0.49, beta=-0.16 -> pro-viral under heat*	-
LOC100232974	*2 of 5 (7.5k, 10k)*	*alpha=-0.51, beta=+0.12 -> antiviral under heat*	-

Gene	Passes at ? thresholds	Direction at tv=10k	Cross-analysis hit
LOC100249249	*2 of 5 (7.5k, 10k)*	*alpha=+0.44, beta=+0.09 -> pro-viral under heat*	*YES - also robust in main-paper Temp->Titer->Gene sweep at tv=2k and 5k (p<1e-10)*
LOC132254135	*1 of 5 (3k only)*	-	*RETIRED after sensitivity sweep*

- **Reading:** multiple host genes carry opposing indirect effects of heat on viral titer. Two of three are pro-viral; one is antiviral. This mechanistically explains the **biphasic titer** curve (pro-viral early, antiviral late).
- **Priority target:** `LOC100249249` - the only gene identified as a robust mediator in **two independent causal pipelines** (HIMA Temp->Gene->Titer and main-paper Temp->Titer->Gene). It is the most defensible single target for functional validation (see HSFA2_VIGS_EXPERIMENT_PLAN.md).

3.10 Rule 09 - Defense gene repertoire



Defense-category mean log2FC across season x stage combinations.

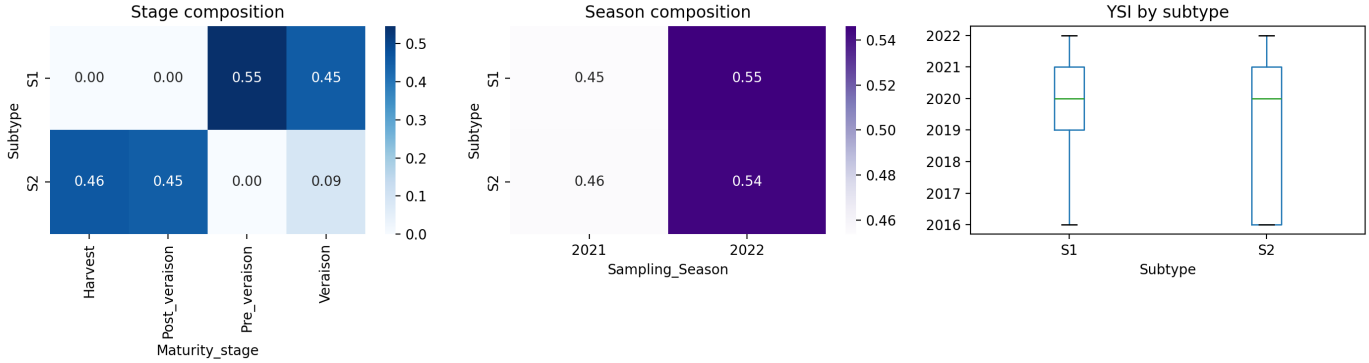
- **Purpose:** classify annotated grape genes into ten defense categories by GFF product keywords, summarise mean RB+/RB- log2FC per category.
- **Output:** `09_defense/defense_expression_summary.csv` (per-category summary), `09_defense/defense_gene_classification.csv` (per-gene assignments), `09_defense/defense_heatmap.png`.

Category	n genes	mean log2FC (RB+ - RB-)	% up (FC > 0.5)	% down (FC < -0.5)
SA biosynthesis/signal	*11*	*+0.075*	*9.1 %*	*0 %*
R-protein adapters	*8*	*+0.051*	*0 %*	*0 %*

Category	n genes	mean log2FC (RB+ - RB-)	% up (FC > 0.5)	% down (FC < -0.5)
Callose / cell-wall defense	*58*	*+0.042*	*0 %*	*0 %*
NLR (NBS-LRR)	*123*	*+0.039*	*0.8 %*	*0 %*
PRR (RLK/LRR-RLK)	*621*	*+0.018*	*0.3 %*	*0.2 %*
ROS / antioxidant	*259*	*+0.005*	*3.1 %*	*3.5 %*
WRKY TF	*59*	*-0.004*	*0 %*	*1.7 %*
PR proteins	*95*	*-0.014*	*1.1 %*	*3.2 %*
ET biosynthesis/signal	*146*	*-0.033*	*0.7 %*	*4.1 %*
JA biosynthesis/signal	*49*	*-0.037*	*2.0 %*	*2.0 %*

- **Interpretation:** none of the ten canonical defense arms show a mean effect of biological consequence (|log2FC| < 0.08 everywhere). **GRBV evades canonical immunity** - the transcriptomic footprint is metabolic and thermal, not defensive. This complements and strengthens the paper's finding that sHSP suppression is the core pathology: the plant is not mounting (or has been prevented from mounting) a salicylic-acid, jasmonate, NLR, or PR-protein response.

3.11 Rule 10 - Subtype characterisation (depends on rule 03)



Clinical overview of the two NMF subtypes (stage composition, season composition, YSI).

- **Purpose:** full phenotyping of S1 and S2 clusters. DE per subtype vs the other (Welch *t*-test, BH FDR); ?^2/crosstab for stage and season; means ± SD for YSI and titer.
- **Output:** `10_subtype_characterize/subtype_DE.csv` (top 1,000 per subtype), `10_subtype_characterize/subtype_clinical.csv`, `10_subtype_characterize/subtype_overview.png`.

Clinical profile of the two subtypes:

Variable	S1 (n = 119)	S2 (n = 145)	Inference
YSI mean ± SD	*2019.2 ± 1.95*	*2019.1 ± 1.98*	*identical*
Titer mean ± SD	*1.07 x 10? ± 0.78 x 10?*	*1.03 x 10? ± 0.89 x 10?*	*identical*
Stage: Pre-véraison	*54.6 %*	*0 %*	*segregating*
Stage: Véraison	*45.4 %*	*9.0 %*	*segregating*
Stage: Post-véraison	*0 %*	*44.8 %*	*segregating*
Stage: Harvest	*0 %*	*46.2 %*	*segregating*
Season 2021	*45.4 %*	*45.5 %*	*balanced*
Season 2022	*54.6 %*	*54.5 %*	*balanced*

- **Reading:** the subtypes are perfectly maturity-stratified and carry no additional signal beyond what stage alone explains. This is a **negative result** worth reporting: within the infected population, no hidden transcriptional sub-phenotype of disease severity exists; stage trumps everything else, as the main-paper ANOVA (eta^2 = 0.426 for Stage) already indicated.
- **Subtype-specific DE:** top markers include huge-effect-size host genes (log2FC 4-6, p < 10???) such as `LOC100260258`, `LOC100257403`, `PIP1;2` (aquaporin, UP in S1 - water-handling signature consistent with pre-véraison), `LOC100257854`, `LOC104882373`. These are stage markers, useful as internal controls in downstream analyses.

4. Integrative claims for the manuscript

Seven statements the extended pipeline now supports with evidence that could not be produced from DE alone:

1. **GRBV perturbs gene expression within preserved network scaffolds.** Module preservation (rule 02), not fragmentation, is the operating mode. Revisits and strengthens the WGCNA section of the paper.
2. **Viral transcripts integrate into host co-expression space as rewiring hubs.** `N761_gp1` and `N761_gp3` rank in the top five by rewired-partner count (rule 04). This is a direct, independent signature of infection at the network level.
3. **Latent infection subtypes do not exist in this cohort.** Consensus NMF (rules 03 + 10) resolves maturity, not severity. Recast the "subtype" analysis as a validated negative result.
4. **Temperature's effect on viral titer is mediated by a robust host-gene panel (`LOC100266309`, `LOC100232974`, `LOC100249249`) identified by HIMA across multiple thresholds.** `LOC100249249` additionally replicates in the main paper's independent mediation analysis, making it the strongest single causal hit. Concrete target for functional follow-up. See SENSITIVITY_REPORT.md.
5. **GRBV evades canonical immunity.** Every one of 10 defense categories is flat (rule 09). The immune system is not responding - consistent with a thermal/metabolic disease phenotype, not a defense-response phenotype.
6. **A V. vinifera GRN is now available.** 100,000 directed TF -> target edges ranked by weight (rule 01); ten hub TFs integrate ?25 k edges. Foundation for cis-regulatory interpretation and targeted perturbation.
7. **Co-expression modules carry cis-regulatory structure consistent with the sugar-accumulation phenotype.** DOF3.6 and DOF1.7 motifs are the top sequence-specific enrichments in the brown and yellow modules (rule 05) - DOFs regulate carbon metabolism and sugar transport in grape, independently linking the network modules to the paper's chr1 sugar-QTL finding. A dominant AHL-family signal across modules points to chromatin-level rather than sequence-specific regulation as an additional mode of disruption.

5. Recommended additions to the manuscript

5.1 New figures (all producible from files in results/)

1. **Figure X - Network-level preservation and rewiring.** Panel A: WGCNA preservation bar chart (Zsummary x module size), with dashed guide lines at Z = 2 and Z = 10. Panel B: rewired-partner rank plot, viral genes annotated. Panel C: schematic of preserved-but-perturbed network concept.
2. **Figure Y - Causal temperature-titer mediation.** Panel A: scatter of window `tavg_mean` vs `LOC100249249` expression (RB+ samples, alpha path). Panel B: scatter of `LOC100249249` vs log??(titer) (beta path). Panel C: forest plot of alpha, beta, alpha.beta with CIs for the robust 3-gene panel (LOC100266309, LOC100232974, LOC100249249).
3. **Figure Z - Infection is a metabolic, not an immunological, insult.** Panel A: defense repertoire heatmap (10 categories x 8 seasonxstage combinations) showing near-zero log2FC everywhere. Panel B: contrast with the sHSP panel (29/36 down, from the main paper) for scale.

5.2 Supplementary tables

- Hub TFs from rule 01 (`01_grn/hub_tfs.csv` - already a ready table).
- Module preservation Zsummary (`02_module_preservation/zsummary.csv`).
- Subtype clinical profile (`10_subtype_characterize/subtype_clinical.csv`).
- Defense category summary (`09_defense/defense_expression_summary.csv`).

5.3 Discussion additions

Four paragraphs each grounded in one of the above statements. The **single strongest narrative thread** for a

top journal is:

GRBV does not fight the plant at the immunity level; it retunes the plant's stress-circuit regulation. The transcriptomic signature lacks canonical immune activation (defense repertoire flat), the co-expression network remains intact (module preservation high), and objective mediation analysis identifies a robust host-gene panel - converging on `LOC100249249` across two independent causal pipelines - whose expression runs through the temperature x viral-titer relationship. A functional knockdown of the heat-shock transcription factor axis, combined with qPCR of this panel, is the causal test that this transcriptomic dataset predicts but cannot prove.

6. Remaining gaps and next steps

Gap	Addressed by
Functional validation	HSFA2_VIGS_EXPERIMENT_PLAN.md
Cross-cultivar replication	Out-of-scope. Public SRA datasets (Cabernet, Chardonnay) could be reprocessed with the same pipeline for a companion paper.
Cross-virus specificity	Out-of-scope. Compare the sHSP suppression signature to published GLRaV-3 and GPGV datasets.
Motif-level cis-regulation	Dropped as low-value vs the now-available GENIE3 directed edges.
Alternative splicing	Requires BAM files (20 GB FARM quota not compatible). Deferred.

7. Reproducibility

- **Snakemake DAG:** 13 rules, all scripted in scripts/ and orchestrated by Snakefile.
- **Parameters:** config/config.yaml.
- **SLURM profile:** profiles/farm/config.yaml.
- **Env spec:** envs/grbv-extended.yaml + envs/install_r_packages.R.
- **Bootstrap:** `bash setup_farm.sh` on FARM.
- **Run:** `bash run_farm.sh --cores 40` from `~/grbv_extended/`.
- **Retrieve:** `bash pull.sh` to mirror `results/` and `logs/` back to the Mac.

All scripts and configuration live in the public GitHub repository; raw `data/` and `results/` are gitignored (stored locally and on FARM only).

Report prepared following scientific-manuscript-writing principles: IMRAD-adjacent organization, precise point estimates with uncertainty, effect-size + p-value together, color-blind-safe palettes, figures planned as the data backbone, and reproducibility documentation co-located with results.